BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: MRD XI 21(A) [INCOMPLETE: Subjective not found]

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** MRD XI 21(A)

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading. [INCOMPLETE PAPER: Subjective Sections B & C not found on any source — objective only.]

Q1. Choose the correct option i.e A, B, C or D.

Q1. Efficiency of a chemical reaction can be checked by calculating the.....

(A) Actual yield
(B) Theoretical yield
(C) Percentage yield
(D) Amount of reactant unused

Q2. The maximum number of electrons in 3S orbital is.....

(A) 1
(B) 2
(C) 3
(D) 6

Q3. The splitting of spectral lines by the application of magnetic field is....

(A) Zeeman effect
(B) Stark effect
(C) Aufbau Principle
(D) Pauli Exclusion Principle

Q4. The bond angle in Ammonia (NH_3) is.....

(A) 90°
(B) 109.5°
(C) 107.5°
(D) 104.5°

Q5. Which one of the given is paramagnetic molecule?

(A) H_2
(B) O_2
(C) N_2
(D) He

Q6. The value of Absolute Zero temperature is equal to.....

(A) −273.15 k
(B) −273.15°C
(C) 0°C
(D) 273.15°C

Q7. The value of universal gas constant (R) at STP is..... atm.dm3.mol^{-1}k^{-1}.

(A) 0.0821
(B) 8.313
(C) 1.987
(D) 62.363

Q8. The steps of a double-helix ladder of DNA are formed by.....

(A) Dipole-Dipole forces
(B) Hydrogen bonds
(C) London Dispersion forces
(D) Ion-Dipole forces

Q9. .....are anisotropic in nature.

(A) Crystalline solids
(B) Non crystalline solids
(C) Liquids
(D) Gases

Q10. Which of the given does not affect the equilibrium constant value of a reaction?

(A) Concentration
(B) Pressure
(C) Temperature
(D) Catalyst

Q11. The value of K_p is greater than K_c when the difference of the moles of the products and reactants is.....

(A) Zero
(B) Positive
(C) Negative
(D) One

Q12. Very large K_a value means that the substance is.....

(A) Strong acid
(B) Weak acid
(C) Neutral
(D) Strong base

Q13. As the number of effective collisions between the reacting particles increases, the rate of reaction.....

(A) Decreases
(B) Increases
(C) Remains the same
(D) Reduces

Q14. The sum of mole fractions of two components of a solution is equal to.....

(A) Zero
(B) 1
(C) 2
(D) 3

Q15. The term cryoscopic constant is related with.....

(A) Boiling point elevation
(B) Freezing point depression
(C) Lowering of vapour pressure
(D) Osmotic pressure

Q16. Processes involving solids and liquids have their....

(A) $\Delta H = \Delta E$

(B) $\Delta H > \Delta E$

(C) $\Delta H < \Delta E$

(D) $\Delta E = 0$

Q17. The oxidation number of Cl in $HClO_3$ is.....

(A) -1

(B) $+1$

(C) $+3$

(D) $+5$

Q18. The electrode potential of the standard hydrogen electrode is chosen as.....

(A) 0V

(B) 1V

(C) -1V

(D) 2V